

Necomimi illusion: Generating Ownership of Cat Ears through Haptic Feedback via Hair

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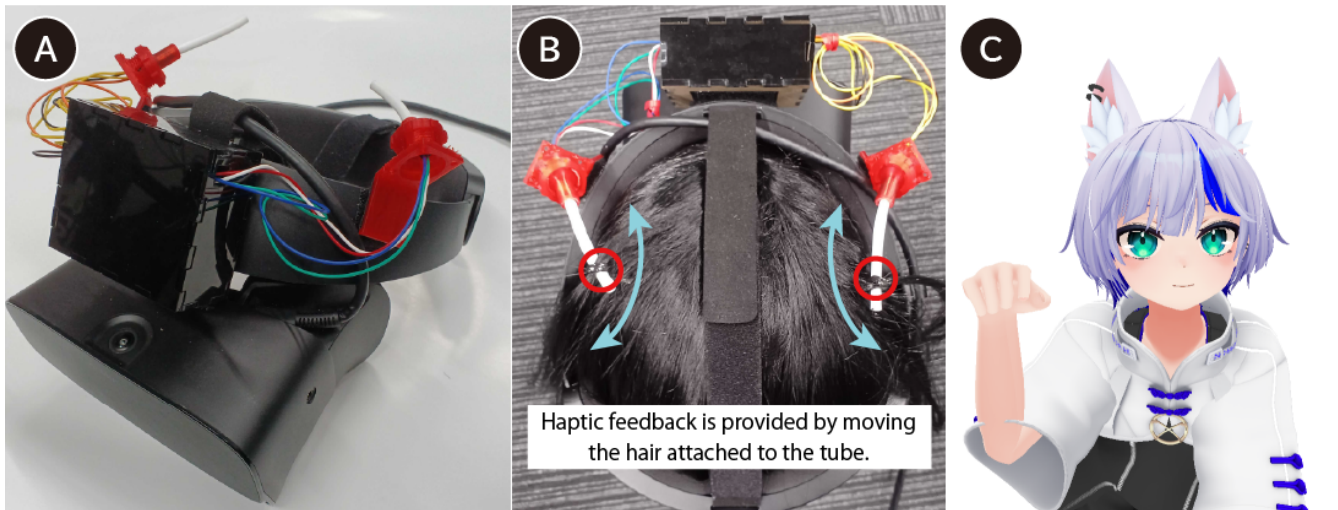


Figure 1: (A): HMD with actuator, (B): Haptic feedback via hair, (C): Avatar with cat ears

Abstract

The sense of ownership over non-human body parts, such as tails and third arms, has been explored using Virtual Reality (VR) technology. These studies are grounded in evolutionary backgrounds and anatomical associations. However, whether the body model can be extended to imaginary body parts without such evolutionary and anatomical associations remains unclear. In this demonstration, we developed a device that creates a sense of ownership over imaginary body parts, specifically cat ears. We achieve this through haptic feedback to the head via hair. Using soft actuators made of shape memory alloys, we move the hair to reproduce various haptic sensations and the comfort associated with stroking cat ears, including the range and direction of the touch. Participants transform into an avatar with cat ears and experience having their imaginary cat ears stroked.

CCS Concepts

• Human-centered computing → Haptic devices.

Keywords

Ownership, Cat ears, Hair, Shape memory alloy

ACM Reference Format:

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1 Introduction

The presentation of synchronized visual and haptic feedback can extend the body model to external objects. Recently, the sense of body ownership of non-human body parts has been investigated using Virtual Reality (VR) technology. For example, it has been reported that adding a third arm or tail to an avatar and presenting synchronized visual and haptic feedback can induce a sense of body ownership over these extended parts [Guterstam et al. 2011; Steptoe et al. 2013]. These studies are based on the evolutionary background of the human coccyx, which is a vestigial tail, and the anatomical associations of arms. In other words, tails and arms are parts that have existed, or currently exist, in the human body through the process of evolution. On the other hand, it is not yet clear whether people can extend their body models to imaginary body parts that lack such evolutionary and anatomical associations.

We decided to endow the avatar with cat ears as imaginary body parts. Characters with cat ears are also very popular on social VR platforms. *Necomimi* is the Japanese word, which means cat ears.

Unique self-expression and non-verbal communication using the appearance and movement of cat ears offer new possibilities for interaction that conventional humanoid avatars do not have. From this perspective, cat ears are an indispensable aspect of investigating avatar embodiment in social VR. In this work, we developed a device that provides haptic feedback when virtual cat ears are moved. We introduce the experience of stroking cat ears by using soft actuators made of shape memory alloys to move the hair.

2 Cat Ears Haptic Feedback

In this work, haptic feedback was presented to the head via hair as a position corresponding to cat ears. To evoke a sense of ownership, the spatial congruence between visual and tactile stimuli is crucial [Guterstam et al. 2011]. Human body hair can detect various haptic information, such as the size of the contacted object, the direction of the haptic sensation and the location, and is also known to induce emotional sensations such as comfort [Boldu et al. 2019]. We reproduced various haptic information and sensations of comfort, such as the range and direction of the haptic sensation when stroking cat ears through the hair.

In this work, we used a soft actuator utilizing shape memory alloy fiber (BMF100¹, Toki Corporation) to provide haptic stimulation. BMF100 is an ultra-thin artificial muscle actuator that contracts and relaxes in response to temperature changes, providing high quietness and lightness, thus enhancing immersion in the VR experience. By threading the shape memory alloy through a soft silicone rubber tube with a diameter of 5mm and a length of 60mm, the entire tube bends when the shape memory alloy contracts and relaxes (Fig. 2). The actuator was controlled using resistive heating of the BMF100, which has a diameter of 0.1mm, a practical force generation of 70gf, and a tensile strength of 0.8Kgf.

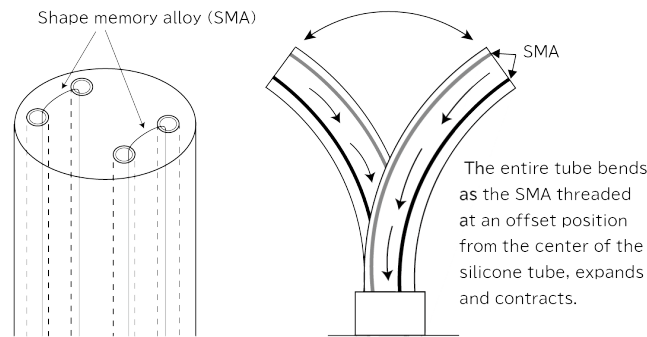


Figure 2: Left: Actuator tip. Right: Mechanism of the actuator movement.

3 Demonstration

Participants wear a head-mounted display (HMD) integrated with actuators. In the VR environment, they transform into an avatar² with anime-style cat ears. Participants observe their avatar from a first-person perspective through a mirror placed in front of them. An anime-style non-player character (NPC) avatar³ is positioned

next to the participants and strokes their cat ears (Fig. 3). This action visually presents the natural movement of the cat ears. The actuators, synchronized with the movement of the cat ears, operate and move the participants' hair to present haptic feedback, allowing them to experience the sensation of having their imaginary cat ears stroked (Fig. 1). The delay between the visual presentation of the cat ears being stroked and the actuators' operation is 63 ms.



Figure 3: The demonstration of stroking cat ears

4 Conclusion

In this demonstration, synchronized visual and haptic feedback provides participants with the sensation of having imaginary cat ears stroked. The haptic feedback through hair replicates various haptic sensations and comfort associated with stroking the avatar's cat ears. Our work allows the body model to extend to the imaginary body part of cat ears and generate a sense of ownership. This study presents new possibilities for avatar interactions that are not present in traditional humanoid avatars.

Acknowledgments

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¹<https://www.toki.co.jp/biometal/products/bmf/bmf.php>

²かなりあ, <https://booth.pm/ja/items/3818504>

³@三丁目の魔界, <https://booth.pm/ja/items/3741802>